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Designing a Web Generator for MSMEs for Business Digitalization

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***Abstract*—Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam sed dapibus erat, nec varius nulla. Ut vitae nunc lorem. Etiam non ornare velit, sed rutrum odio. Vivamus a lacinia velit. Integer fringilla tincidunt rhoncus. Etiam hendrerit pharetra dapibus. Curabitur in luctus enim. Aenean lacus lectus, imperdiet ut lectus sit amet, mollis convallis enim. Ut tempor elit laoreet nulla tempus iaculis. Aliquam erat volutpat. Aliquam nec turpis ut metus venenatis dapibus et in justo.**

***Index Terms*—keyword1, keyword2, keyword3**

I. Introduction

The Indonesian digital economy is currently experiencing a period of rapid growth, with projections estimating that its value will reach USD 150 billion by 2025. Despite this potential, the Micro, Small, and Medium Enterprises (MSME) sector—which serves as the backbone of the national economy by contributing 61% to the GDP—continues to face significant challenges in fully participating in this digital transformation. Research by Rajagukguk and Rusadi (2025) highlights a clear "adoption gap," revealing that while 65% of MSMEs recognize the necessity of digital technology, only 23% are able to utilize it consistently.

This disparity stems largely from the incompatibility of existing technological tools with the needs and capabilities of non-technical MSME owners. Currently, these business owners are often forced into difficult trade-offs: either adopting marketplace platforms that impose administrative fees—averaging 6.5%—which erode thin profit margins and limit control over customer data, or attempting to build independent websites using complex Content Management Systems (CMS) like WordPress, which often prove too technically demanding for those with limited digital literacy. Furthermore, popular international website builders (e.g., Wix) frequently lack native integration with local Indonesian digital habits, such as QRIS payments and automated WhatsApp communication.

To address these barriers, this research proposes the development of a low-code/no-code web generator platform specifically tailored for Indonesian MSMEs. This platform aims to provide a "middle ground" solution that is intuitive for non-technical users, cost-effective by eliminating commission-based dependencies, and locally relevant through built-in support for QRIS and WhatsApp. The objective of this study is to design a functional Minimum Viable Product (MVP) and empirically demonstrate, through System Usability Scale (SUS) benchmarking, that this platform offers a significantly higher level of usability and ease of creation for MSME owners compared to conventional CMS alternatives. By providing a professional yet accessible digital presence, this research contributes to the broader effort of democratizing digital technology for small businesses across Indonesia.

II. Literature Review

A. Digital Transformation of MSMEs

Micro, Small, and Medium Enterprises (MSMEs) represent a fundamental pillar of economic development, particularly in emerging economies, contributing significantly to employment generation, income distribution, and national Gross Domestic Product (GDP) [9]. Despite their strategic importance, MSMEs face substantial challenges in adapting to the rapidly evolving digital economy.

The advancement of digital technologies offers significant opportunities for MSMEs to expand market reach, improve operational efficiency, and enhance customer engagement. However, the adoption of digital tools remains uneven. Prior studies highlight the existence of an "adoption gap," where awareness of digital technologies does not necessarily translate into actual implementation [6]. This gap is primarily influenced by limited digital skills, financial constraints, and lack of technical support.

Furthermore, digital transformation is not solely a technological shift but also involves organizational and cultural changes [1]. For MSMEs with limited resources and informal structures, this transformation becomes significantly more challenging. Therefore, the success of digital transformation initiatives largely depends on the accessibility, affordability, and usability of the provided technologies.

B. Limitations of Existing Digital Platforms

Existing digital solutions for MSMEs can generally be categorized into marketplace platforms and Content Management Systems (CMS). Marketplace platforms offer ease of use and immediate access to customers; however, they impose transaction fees and restrict control over customer data, which may negatively impact long-term business sustainability [6].

Conversely, CMS platforms such as WordPress provide greater flexibility and control but require a certain level of technical expertise. This makes them less suitable for MSME owners with limited digital literacy. Website builders such as Wix attempt to simplify development processes; however, these platforms are typically designed for global markets and often lack support for local business practices.

While these solutions address specific aspects of digitalization, they fail to simultaneously provide ease of use, flexibility, and localization. This limitation indicates the need for a more balanced and accessible approach tailored specifically to MSMEs.

C. Low-Code/No-Code Development

One promising approach to reducing technical barriers is the adoption of low-code and no-code development platforms. These platforms enable users to create applications through visual interfaces with minimal programming knowledge. Research by Sahay et al. demonstrates that low-code platforms significantly accelerate development processes and empower non-programmers to build functional applications [4].

Similarly, Kumar and Pandey found that no-code platforms improve accessibility and reduce development time, particularly for small businesses [5]. Despite these advantages, most existing platforms remain generic and are not specifically designed to address the contextual needs of MSMEs, particularly in developing countries.

Therefore, while low-code/no-code approaches reduce technical complexity, they still require further adaptation to support localized business environments and user-specific requirements.

D. User-Centered Design (UCD)

To address usability challenges, User-Centered Design (UCD) has emerged as a critical approach in system development. UCD emphasizes designing systems based on user needs, capabilities, and limitations, making it particularly relevant for MSMEs with low digital literacy.

Studies indicate that applying UCD principles can significantly improve usability and user satisfaction by reducing system complexity and aligning system functionalities with real-world user behavior [7]. In the context of MSMEs, this approach ensures that digital tools are not only functional but also accessible and intuitive.

However, many existing MSME digital solutions do not fully incorporate UCD principles, resulting in systems that are technically capable but difficult to adopt in practice.

E. Usability Evaluation Using System Usability Scale (SUS)

Usability is a critical factor influencing the successful adoption of digital systems. The System Usability Scale (SUS), introduced by Brooke, is a widely used and reliable tool for evaluating system usability [2]. It provides a simple yet effective way to measure user perception through a standardized questionnaire.

Bangor et al. further validated SUS as a robust and versatile instrument for assessing usability across various systems, including web-based applications [3]. One key advantage of SUS is its ability to produce quantitative scores that allow direct comparison between systems.

Despite its effectiveness, relatively few studies on MSME digital solutions employ standardized usability metrics such as SUS, limiting the ability to objectively evaluate and compare system performance.

III. Research Methodology

A. Research Design

This study adopts a **quantitative approach** to evaluate the impact of the proposed web generator system on Micro, Small, and Medium Enterprises (MSMEs). The research focuses on measuring changes in business performance **before and after system implementation**.

A **pre-test and post-test design** is employed to capture differences in key performance indicators. This design is widely used to assess the effectiveness of technological interventions in applied research settings [10]. By comparing results before and after system adoption, this study aims to identify measurable improvements attributable to the developed system.

B. Development Method

This study adopts the **User-Centered Design (UCD)** approach in developing the proposed web generator system. UCD emphasizes designing systems based on user needs, capabilities, and limitations, making it particularly suitable for MSMEs with varying levels of digital literacy [11].

The development process consists of **requirement gathering, system design, prototype development, and user evaluation**. Initially, user requirements are collected through observations and informal interviews with MSME owners. Based on these findings, the system is designed to prioritize **usability, simplicity, and relevance**. A prototype is then developed using a **low-code/no-code approach** to accelerate development and reduce technical barriers. Finally, the system is iteratively refined based on user feedback to ensure usability and effectiveness [11].

C. Evaluation Indicators

To evaluate the system's impact, this study uses three main indicators: **content management efficiency**, **market reach expansion**, and **digital visibility improvement**. Content management efficiency measures the time and effort required to manage website content, as digital tools improve operational efficiency [12]. Market reach expansion evaluates the ability of MSMEs to access broader markets beyond their local area [13]. Digital visibility improvement assesses the online presence and discoverability of MSMEs, which contributes to improved business performance [14].

D. Data Collection Method

Data are collected using a **structured questionnaire** distributed to MSME participants in two stages: **pre-test** and **post-test**. The questionnaire uses a **Likert scale (1–5)** to measure user perceptions, where 1 indicates "strongly disagree" and 5 indicates "strongly agree" [15].

E. Data Analysis Technique

The collected data are analyzed using **descriptive and comparative methods**. Descriptive analysis is used to summarize general trends, while comparative analysis examines differences between pre-test and post-test results to determine system effectiveness [16].

F. Research Procedure

The research procedure includes **participant identification, pre-test data collection, system implementation, post-test data collection, and data analysis**.

IV. Results and Discussion

Tampilkan hasil dan pembahasan.

V. Conclusion

Tulis kesimpulan.

Acknowledgment

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